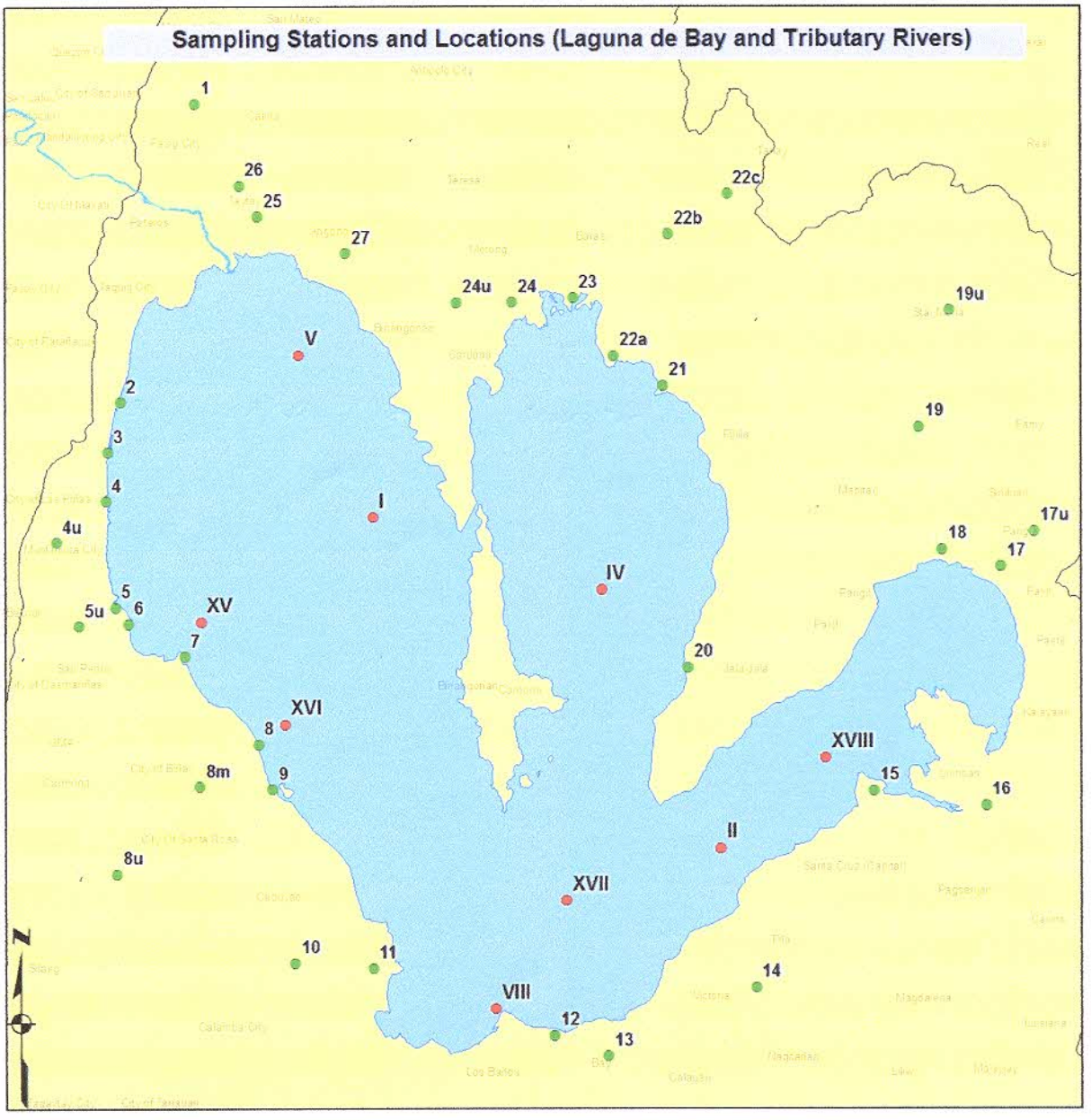
LLDA Quarterly Water Quality Monitoring Report
Laguna Lake and Tributary Rivers
July to September, 2016



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LLDA Quarterly Water Quality Report
Laguna Lake and Tributary Rivers
July to September, 2016

Laguna Lake Stations

Monitoring Stations	Water Quality Parameters*								
	BOD (mg/l)			Dissolved Oxygen (mg/l)			Total coliforms (MPN/100ml)**		
	July	Aug	Sept	July	Aug	Sept	July	Aug	Sept
Stn. I (Central West Bay)	2	3	3	7.3	5.5	7.3	185	126	335
Stn. II (East Bay)	1	2	2	9.7	6.8	9.0	586	665	754
Stn. IV (Central Bay)	6	2	3	15.9	7.3	13.5	104	121	49
Stn. V (Northern West Bay)	2	2	2	6.3	6.6	7.8	793	945	305
Stn. VIII (South Bay)	1	2	2	8.1	7.6	8.8	517	586	272
Stn.XV (San Pedro)	2	2	3	4.0	6.5	8.7	786	418	418
Stn.XVI (Sta. Rosa)	*2*	*7*	*4*	*7.5*	*7.4*	*10.2*	*136*	*211*	*211*
Stn.XVII (Sanctuary)	2	2	3	7.8	7.3	12.1	41	46	82
Stn.XVIII (Pagsanjan)	3	3	2	9.4	6.7	9.4	38	99	194

Notes: * DENR Class C Water Quality Criteria:
Biochemical Oxygen Demand (BOD) - 10 mg/L
Dissolved Oxygen (DO) - 5 mg/L
Total Coliforms - 5000 MPN/100 ml
** Based on computed geomean

Tributaty Rivers Stations

Station	Monitoring Stations	Water Quality Parameters*								
		BOD (mg/l)			Dissolved Oxygen (mg/L)			Total coliforms (MPN/100ml)**		
		July	Aug	Sept	July	Aug	Sept	July	Aug	Sept
1	Marikina River	26	4	7	0.70	5.10	3.30	1,817,373	663,088	344,308
2	Bagumbayan River (Taguig)	166	79	105	0.05	0.05	0.05	15,303,449	17,518,077	26,277,115
3	Buli Creek (Taguig)	201	109	127	0.05	0.05	0.05	19,957,242	25,906,965	24,624,247
4	Mangangate River (Muntinlupa) - Downstream	74	45	24	0.05	0.05	0.05	11,177,719	4,931,355	7,366,466
4U	Mangangate River (Muntinlupa) - Upstream	8	5	11	5.00	3.50	6.60	729,413	729,413	1,404,746
5	Tunasan River (Muntinlupa) - Downstream	281	144	129	0.05	0.05	0.05	68,671,547	36,487,204	23,313,907
5U	Tunasan River (Muntinlupa) - Upstream	5	12	-	5.50	5.20	-	286,923	259,422	148,661
6	San Pedro River (T_2)	9	10	16	0.05	0.05	2.30	659,444	1,083,074	793,965
7	Biñan River	22	60	21	1.90	0.05	1.20	5,122,317	5,918,909	3,054,986
8	Sta. Rosa River - Downstream	55	25	26	0.05	0.05	0.90	8,501,269	13,304,828	10,515,864
8M	Sta. Rosa River - Midstream	18	42	31	0.05	0.05	0.90	6,712,134	5,581,486	4,137,705
8U	Sta. Rosa River - Upstream	5	7	12	6.10	5.80	7.70	488,103	586,979	1,113,981
9	Cabuyao River	54	24	35	0.05	0.05	0.05	16,000,000	11,139,813	9,263,331
10	San Cristobal River (T_3)	104	133	79	0.05	0.05	0.60	30,863,780	42,594,714	80,166,321
11	San Juan River (T_5)	4	5	6	5.10	4.70	5.30	626,990	511,098	426,791
12	Molawin Creek (Los Baños)	6	4	4	3.30	2.00	5.30	59,014	98,063	149,572
13	Bay River (T_9)	4	3	4	5.80	6.40	5.80	1,009,771	1,127,987	582,199
14	Pila River	3	2	3	2.10	4.00	4.50	52,718	52,718	115,199
15	Sta. Cruz River (T_6)	3	3	4	4.60	6.00	4.00	110,653	110,653	110,653
16	Pagsanjan River (T_8)	2	2	2	6.00	6.10	5.90	139,162	220,905	194,799
17	Pangil River - Downstream	2	1	5	8.60	7.20	5.50	327,977	257,812	456,610
17U	Pangil River - Upstream	1	1	1	7.70	7.80	7.70	10,591	10,591	18,292
18	Siniloan River	2	3	8	4.10	4.70	2.80	237,095	216,814	254,232
19	*Sta. Maria River - Downstream*	*2*	*2*	*3*	*5.30*	*6.40*	*6.70*	*219,848*	*193,867*	*587,844*
19U	Sta. Maria River - Upstream	2	3	2	8.70	7.60	7.90	36,487	43,878	160,000
20	Jala-jala River	-	-	-	-	-	-	-	-	-
21	Pililla River	3	3	4	1.40	6.80	4.40	926,333	630,162	630,162
22 A	Tanay River - Brgy. Wawa	5	4	6	1.80	6.00	2.00	20,263	46,784	344,710
22 B	Tanay River - Daranak Falls	1	2	1	8.20	8.90	0.50	9,327	21,534	58,698
22 C	Tanay River- Cuyambay	1	4	-	7.30	7.00	-	20,770	44,748	74,833
23	Baras River	7	6	5	2.10	5.10	4.90	335,926	412,097	591,891
24	Morong River - Downstream	16	9	7	3.90	4.20	7.50	240,000	165,008	162,684
24U	Morong River - Upstream	14	3	6	3.80	4.60	4.90	59,459	79,541	64,710
25	Manggahan Floodway (Taytay)	-	3	11	-	4.90	3.10	742,967	328,634	222,486
26	*Sapang Baho River (Cainta)*	*14*	*15*	*15*	*0.50*	*0.80*	*0.60*	*4,259,471*	*2,593,435*	*1,879,182*
27	Angono River	13	3	8	4.50	5.10	4.60	2,089,746	2,089,746	2,907,156

Notes: * DENR Class C Water Quality Criteria:
Biochemical Oxygen Demand (BOD) - 10 mg/L
Dissolved Oxygen (DO) - 5 mg/L
Total Coliforms - 5000 MPN/100 ml
** Based on computed geomean
*** Stations renamed were Tanay Upstream to Tanay River-Daranak Falls and Tanay Downstream to Tanay River - Brgy. Wawa after Tanay - Cuyambay monitoring station was added starting in February 2016

LLDA Quarterly Water Quality Report
Laguna Lake and Tributary Rivers
July to September, 2016

Laguna Lake Stations

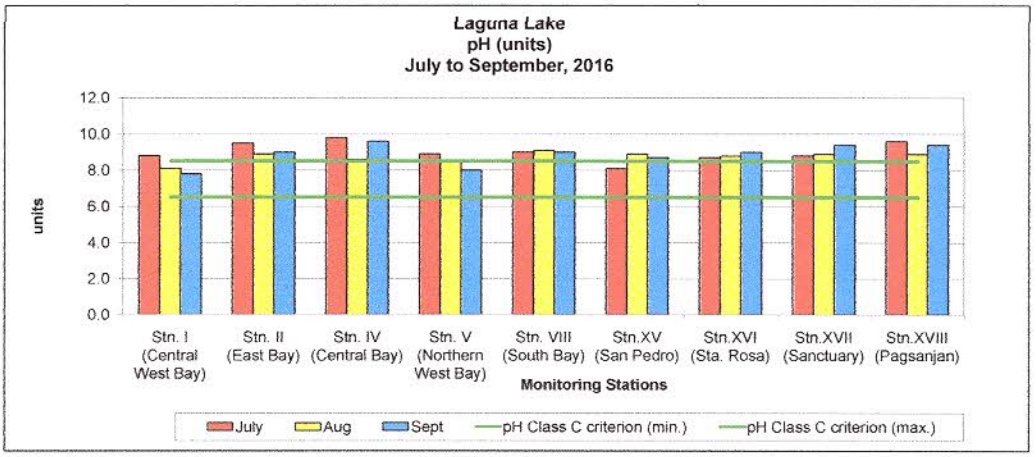
Monitoring Stations	Water Quality Parameters											
	pH (units)			Ammonia (mg/L)			Nitrate (mg/L)			Inorganic Phosphate (mg/L)		
	July	Aug	Sept	July	Aug	Sept	July	Aug	Sept	July	Aug	Sept
Stn. I (Central West Bay)	8.8	8.1	7.8	0.025	0.313	0.076	0.068	0.382	0.728	0.134	0.235	0.134
Stn. II (East Bay)	9.5	8.9	9.0	0.015	0.181	0.048	0.574	0.271	0.022	0.179	0.001	0.011
Stn. IV (Central Bay)	9.8	8.6	9.6	0.034	0.183	0.073	0.031	0.142	0.058	0.147	0.014	0.039
Stn. V (Northern West Bay)	8.9	8.5	8.0	0.067	0.120	0.029	0.110	0.313	1.084	0.048	0.306	0.145
Stn. VIII (South Bay)	9.0	9.1	9.0	0.031	0.069	0.001	0.001	0.028	0.109	0.143	0.001	0.045
Stn.XV (San Pedro)	8.1	8.9	8.7	0.073	0.070	0.001	0.001	0.301	0.069	0.176	0.008	0.072
Stn.XVI (Sta. Rosa)	8.7	8.8	9.0	0.039	0.033	0.011	0.001	0.001	0.037	0.074	0.009	0.036
Stn.XVII (Sanctuary)	8.8	8.9	9.4	0.028	0.025	0.005	0.001	0.037	0.028	0.134	0.005	0.061
Stn.XVIII (Pagsanjan)	9.6	8.9	9.4	0.001	0.256	0.013	0.038	0.041	0.031	0.001	0.001	0.007

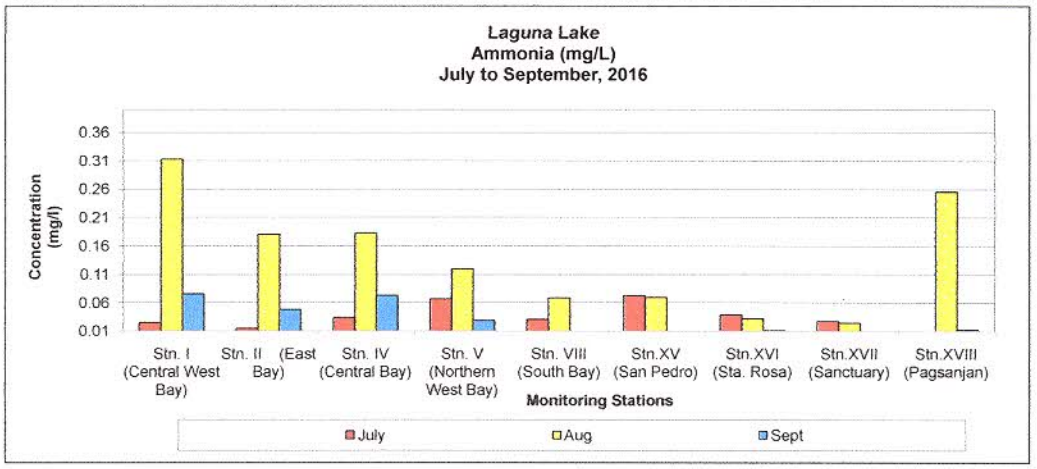
Tributary River Stations

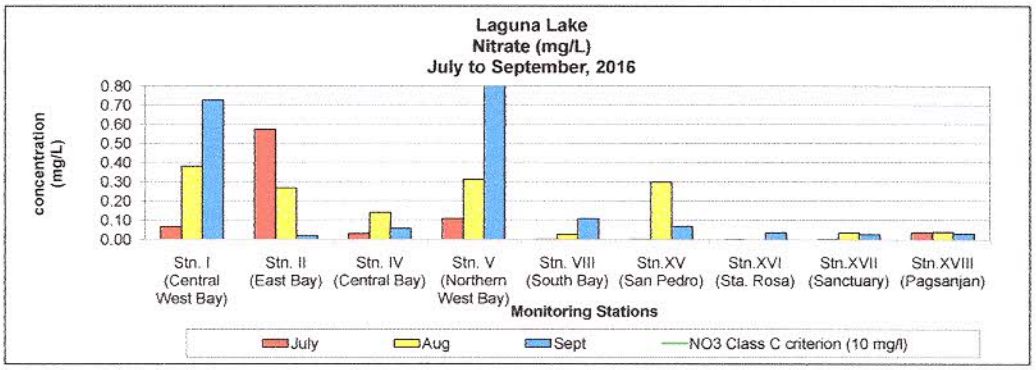
Station	Monitoring Stations	Water Quality Parameters								
		pH (units)			Nitrate (mg/L)			Inorganic Phosphate (mg/L)		
		July	Aug	Sept	July	Aug	Sept	July	Aug	Sept
1	Marikina River	7.6	7.8	7.7	0.557	0.458	0.976	0.169	0.123	0.251
2	Bagumbayan River (Taguig)	7.2	7.4	7.3	0.083	0.264	0.077	2.254	1.610	1.164
3	Buli Creek (Taguig)	7.4	7.5	7.5	0.044	0.105	0.078	1.510	2.050	1.216
4	Mangangate River (Muntinlupa) - Downstream	7.5	7.5	7.3	0.001	0.126	0.083	4.016	1.264	0.638
4U	Mangangate River (Muntinlupa) - Upstream	7.8	7.6	7.6	0.197	0.320	0.423	0.966	0.915	0.239
5	Tunasan River (Muntinlupa) - Downstream	7.0	7.0	7.2	0.020	0.240	0.097	1.038	1.702	0.492
5U	Tunasan River (Muntinlupa) - Upstream	8.1	7.7	-	1.572	3.136	-	1.128	0.851	-
6	San Pedro River (T_2)	7.7	7.7	7.3	1.038	0.605	2.288	0.588	1.514	0.665
7	Biñan River	7.8	7.7	7.4	0.041	0.084	0.045	0.220	1.752	0.492
8	Sta. Rosa River - Downstream	7.8	7.6	7.5	0.023	0.069	0.125	0.220	0.926	0.339
8M	Sta. Rosa River - Midstream	7.8	7.7	7.5	0.096	0.402	0.116	0.160	0.927	0.462
8U	Sta. Rosa River - Upstream	8.4	7.9	7.6	1.249	1.376	0.993	0.081	0.661	0.382
9	Cabuyao River	7.6	7.5	7.3	0.073	0.057	0.024	0.392	1.416	0.632
10	San Cristobal River (T_3)	6.8	6.1	7.0	0.040	0.900	0.212	0.652	0.392	0.321
11	San Juan River (T_5)	7.5	7.5	7.4	3.033	5.052	1.924	1.131	0.828	0.576
12	Molawin Creek (Los Baños)	7.7	7.2	7.5	0.235	0.738	0.867	0.489	0.491	0.308
13	Bay River (T_9)	8.1	8.1	7.5	0.572	1.115	0.696	0.275	0.492	0.289
14	Pila River	7.6	7.4	7.5	0.267	4.484	0.217	0.089	0.312	0.166
15	Sta. Cruz River (T_6)	7.7	7.5	7.4	4.070	1.143	0.663	0.229	0.370	0.287
16	Pagsanjan River (T_8)	7.9	7.7	7.5	0.172	0.474	0.743	0.083	0.038	0.090
17	Pangil River - Downstream	7.5	7.4	7.4	0.092	0.433	0.148	0.159	0.066	0.117
17U	Pangil River - Upstream	7.4	7.3	7.2	0.092	0.799	0.099	0.061	0.141	0.101
18	Siniloan River	7.2	7.5	7.2	0.090	0.796	0.063	0.391	0.175	0.130
19	Sta. Maria River - Downstream	7.8	7.5	7.7	0.082	1.348	0.267	0.489	0.106	0.107
19U	Sta. Maria River - Upstream	8.6	7.7	7.9	0.068	0.765	0.479	0.110	0.074	0.068
20	Jala-jala River	-	-	-	-	-	-	-	-	-
21	Pililla River	7.3	7.4	6.7	0.390	1.988	0.081	1.021	0.287	0.341
22 A	Tanay River - Brgy. Wawa	7.7	7.6	7.9	0.262	0.913	1.171	0.196	0.157	0.178
22 B	Tanay River - Daranak Falls	8.5	8.1	8.5	0.001	0.645	0.699	0.073	0.100	0.124
22 C	Tanay River- Cuyambay	7.8	7.7	-	0.406	0.489	-	0.018	0.002	-
23	Baras River	7.5	8.0	7.5	0.722	2.804	1.900	0.597	0.603	0.483
24	Morong River - Downstream	7.9	6.8	8.2	1.686	1.428	2.542	0.609	1.006	1.708
24U	Morong River - Upstream	7.8	7.3	7.9	1.536	1.642	1.866	0.619	0.904	1.559
25	Manggahan Floodway (Taytay)	-	7.3	7.5	-	0.444	0.059	-	0.085	0.541
26	Sapang Baho River (Cainta)	7.4	7.8	7.5	0.566	1.264	0.072	0.593	0.446	1.000
27	Angono River	7.6	6.4	7.6	2.067	1.852	1.307	0.321	0.312	0.410

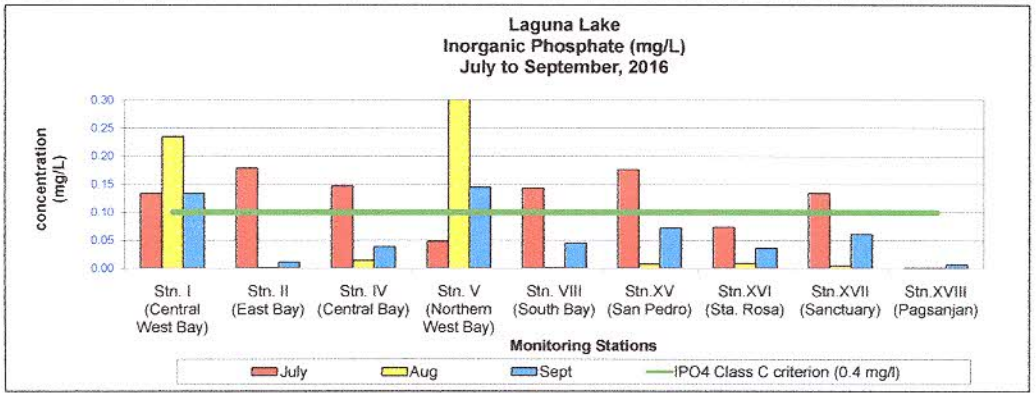
Graphs:

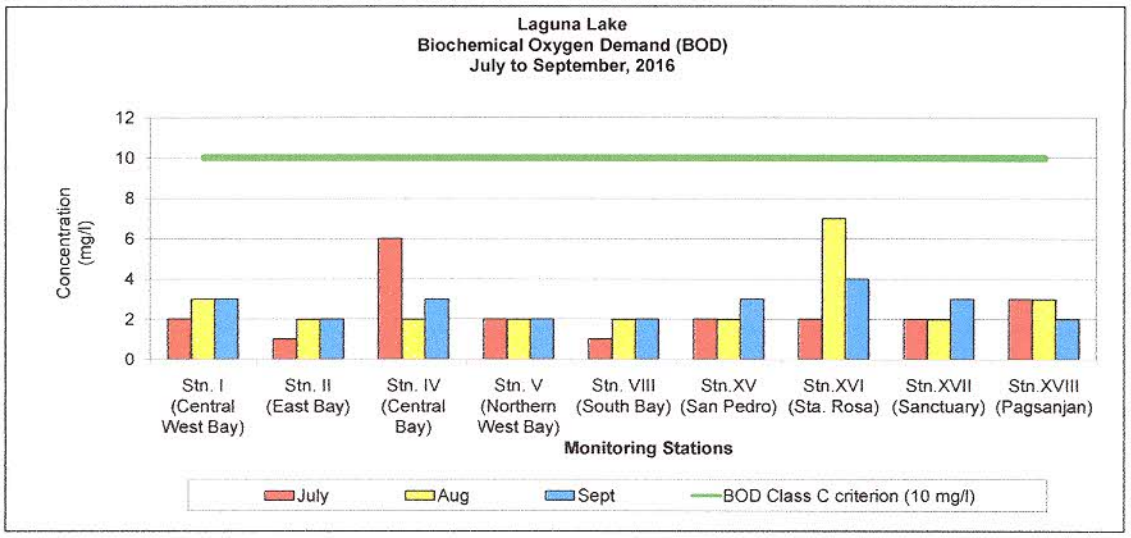
Laguna Lake

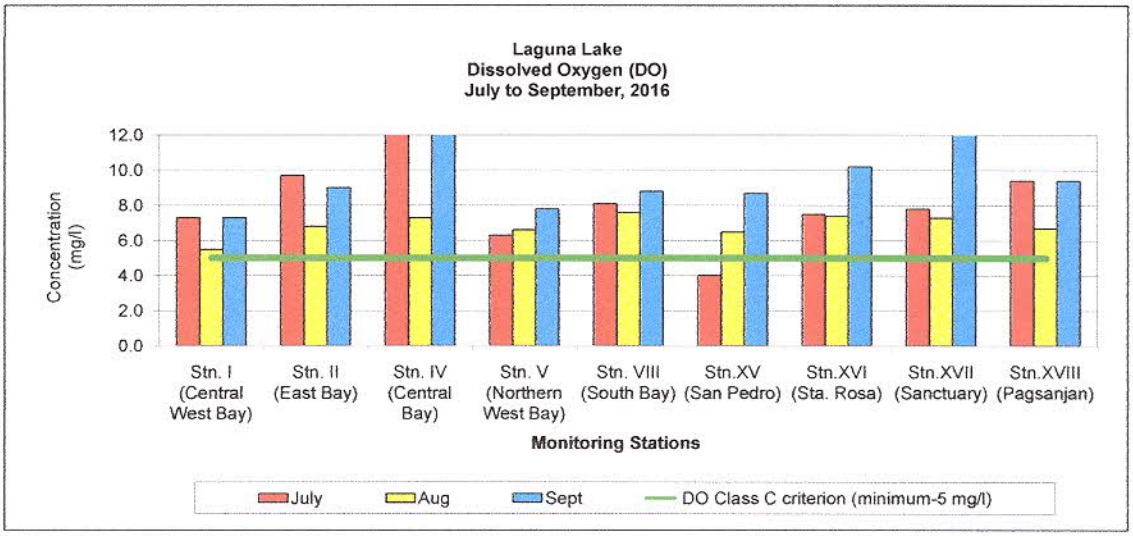


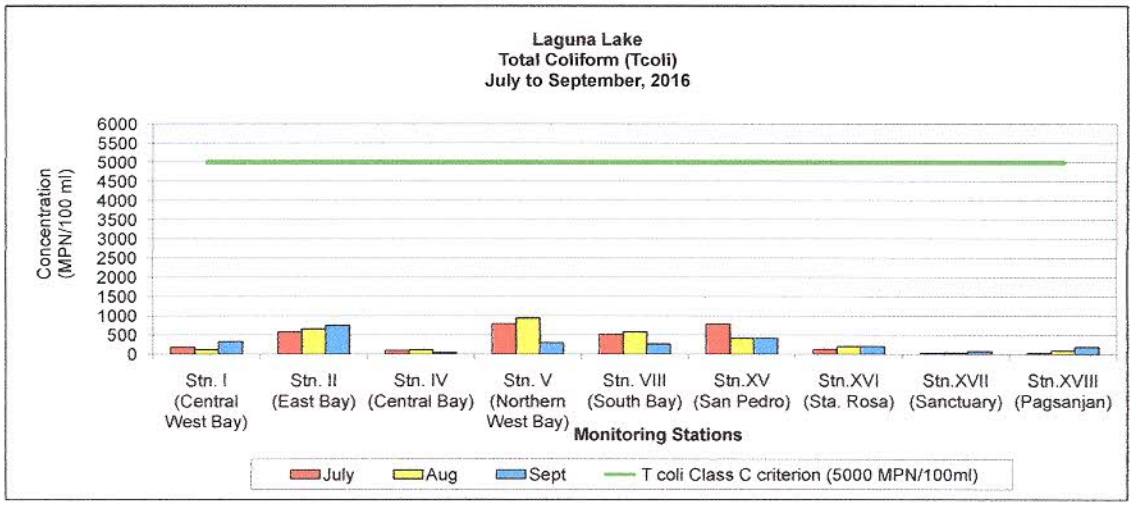




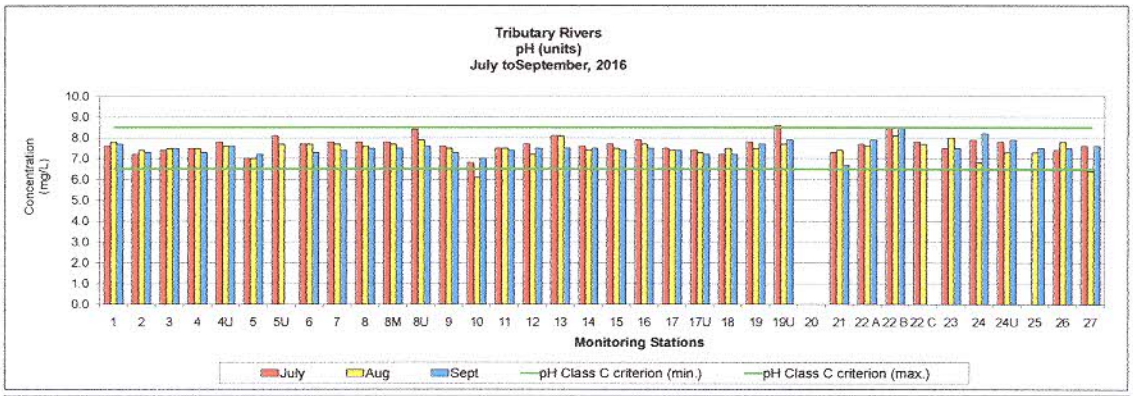


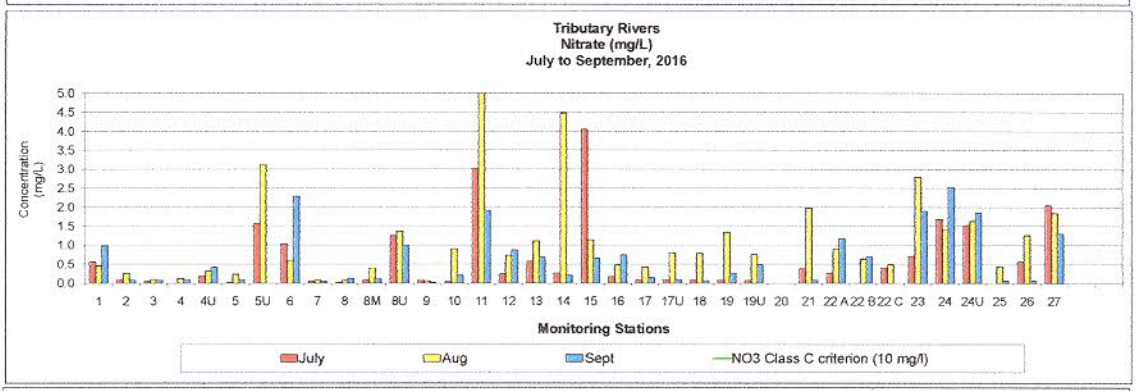

Laguna Lake
Biochemical Oxygen Demand (BOD)
July to September, 2016
Concentration (mg/l)
12
10
8
6
4
2
0
Stn. I (Central West Bay)
Stn. II (East Bay)
Stn. IV (Central Bay)
Stn. V (Northern West Bay)
Stn. VIII (South Bay)
Stn.XV (San Pedro)
Stn.XVI (Sta. Rosa)
Stn.XVII (Sanctuary)
Stn.XVIII (Pagsanjan)
Monitoring Stations
July
Aug
Sept
BOD Class C criterion (10 mg/l)

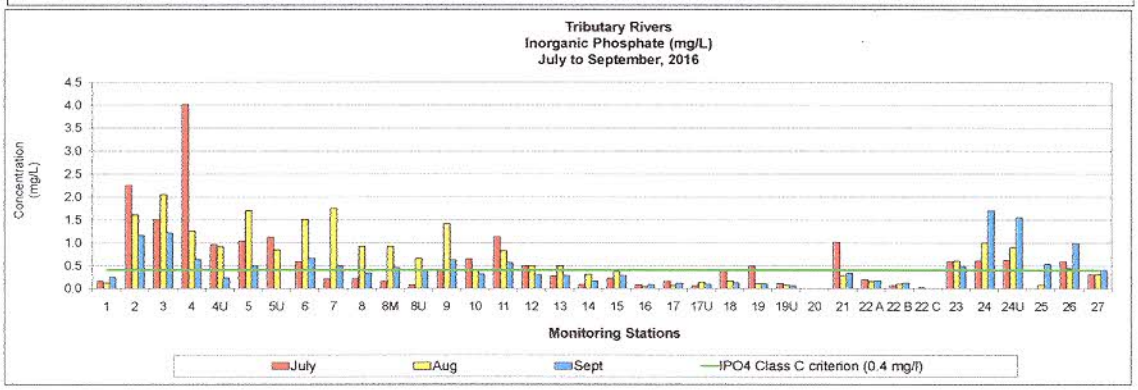

Laguna Lake
Dissolved Oxygen (DO)
July to September, 2016
Concentration (mg/l)
12.0
10.0
8.0
6.0
4.0
2.0
0.0
Stn. I (Central West Bay)
Stn. II (East Bay)
Stn. IV (Central Bay)
Stn. V (Northern West Bay)
Stn. VIII (South Bay)
Stn.XV (San Pedro)
Stn.XVI (Sta. Rosa)
Stn.XVII (Sanctuary)
Stn.XVIII (Pagsanjan)
Monitoring Stations
July
Aug
Sept
DO Class C criterion (minimum-5 mg/l)


Laguna Lake
Total Coliform (Tcoli)
July to September, 2016
Concentration (MPN/100 ml)
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
0
Stn. I (Central West Bay)
Stn. II (East Bay)
Stn. IV (Central Bay)
Stn. V (Northern West Bay)
Stn. VIII (South Bay)
Stn.XV (San Pedro)
Stn.XVI (Sta. Rosa)
Stn.XVII (Sanctuary)
Stn.XVIII (Pagsanjan)
Monitoring Stations
July
Aug
Sept
T coli Class C criterion (5000 MPN/100ml)

Tributary Rivers



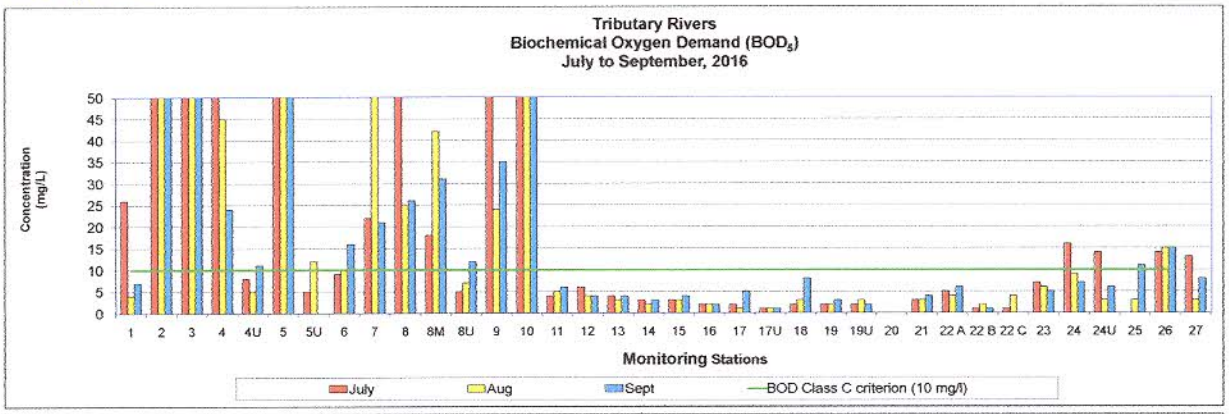


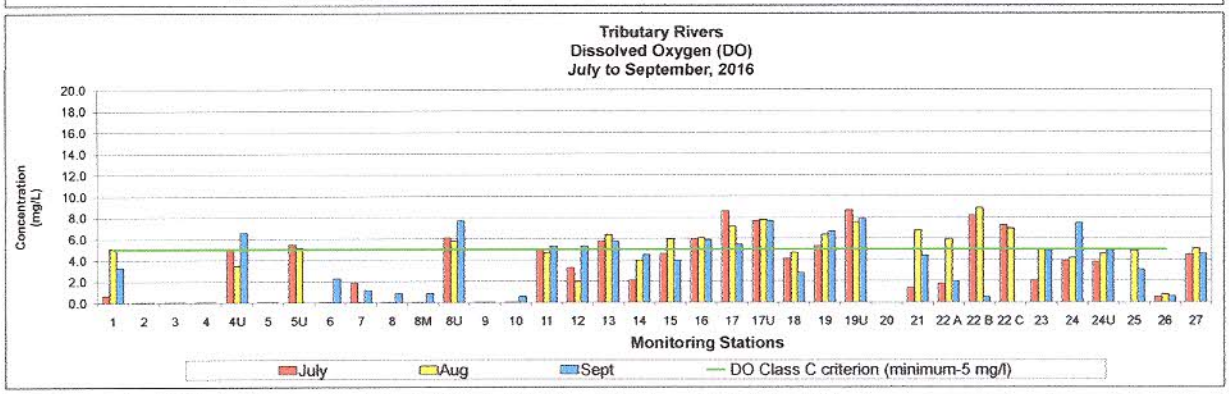


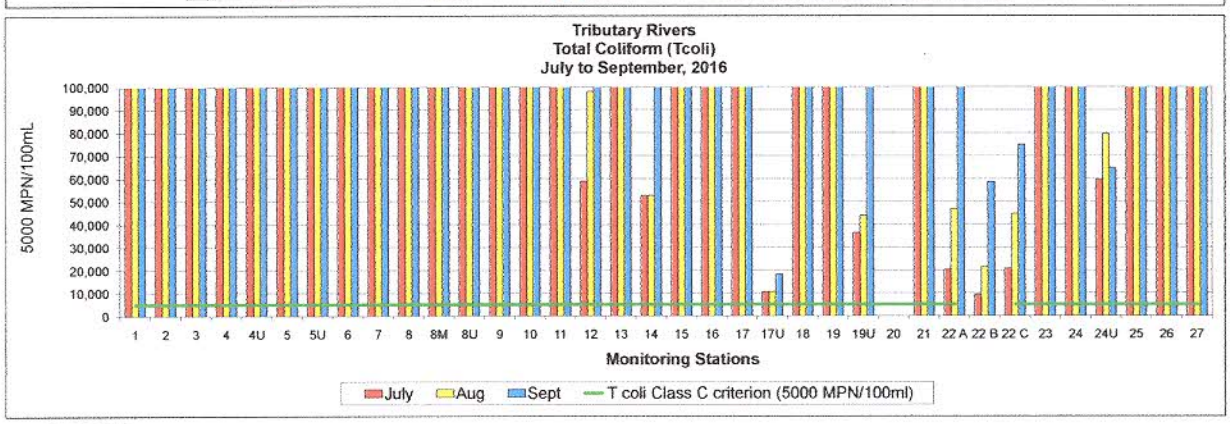
STATIONS

1 *Marikina River*
2 Bagumbayan River(Taguig)
3 Buli Creek (Taguig)
4 Mangangate River (Muntinlupa)-Downstream
4U Mangangate River(Muntinlupa)-Upstream
5 Tunasan River (Muntinlupa)- Downstream
5U Tunasan River(Muntinlupa)- Upstream
6 San Pedro River (T2)
7 Biñan River
8 Sta. Rosa River- Downstream
8M Sta. Rosa River- Midstream
8U Sta. Rosa River- Upstream
9 Cabuyao River
10 San Cristobal River (T3)
11 San Juan River (T5)
12 Los Baños River
13 Bay River (T9)
14 Pila River
15 Sta. Cruz River (T6)
16 *Pagsanjan River (T8)*
17 Pangil River- Downstream
17U Pangil- Upstream
18 Siniloan River
19 Sta. Maria River- Downstream
19U Sta. Maria River- Upstream
20 Jala-jala River
21 Pililla River
22A Tanay River- Brgy. Wawa
22B Tanay River- Daranak Falls
22C Tanay River- Cuyambay
23 Baras River
24 Morong River- Downstream
24U Morong River- Upstream
25 Manggahan Floodway (Taytay)
26 Sapang Baho River (Cainta)
27 Angono River

Tributary Rivers







STATIONS

1 Marikina River
2 Bagumbayan River (Taguig)
3 Buli Creek (Taguig)
4 Mangangate River (Muntinlupa)-Downstream
4U Mangangate River (Muntinlupa)-Upstream
5 Tunasan (Muntinlupa)- Downstream
5U Tunasan (Muntinlupa)- Upstream
6 San Pedro River (T2)
7 Biñan River
8 Sta. Rosa River- Downstream
8M Sta. Rosa River- Midstream
8U Sta. Rosa River- Upstream
9 Cabuyao River
10 San Cristobal River (T3)
11 San Juan River (T5)
12 Molawin Creek (Los Baños)
13 Bay River (T9)
14 Pila River
15 Sta. Cruz River (T6)
16 Pagsanjan River (T8)
17 Pangil River- Downstream
17U Pangil River- Upstream
18 Siniloan River
19 Sta. Maria River- Downstream
19U Sta. Maria River- Upstream
20 Jala-jala River
21 Pililla River
22A Tanay River-Brgy. Wawa
22B Tanay River- Daranak falls
22C Tanay River- Cuyambay
23 Baras River
24 Morong River- Downstream
24U Morong River- Upstream
25 Manggahan Floodway (Taytay)
26 Sapang Baho River(Cainta)
27 Angono River

FINDINGS:

A. On Biochemical Oxygen Demand (BOD):

Laguna Lake:

- For the 3rd quarter of 2016, all of the nine (9) lake stations passed the 10 mg/L BOD criterion set for Class C waters by the DENR.
- The BOD concentrations for the 3rd quarter ranged from 1 to 7 mg/L.

Tributary Rivers:

- From July to September, the results of the BOD analysis showed that out of the 36 tributary stations monitored, only 17 stations consistently passed the Class C criterion for BOD and these were Stns. 11 to 23.
- All of the monthly BOD concentrations in Stns. 2, 3, 4, 5, 7, 8, 8M, 9, 10 and 26 failed the 10 mg/L Class C criterion.
- The BOD concentrations in Stns. 1, 24 and 27 conformed with the criterion in August and September while Stns. 4U, 6, and 8U passed the criterion in July and August.
- Stns. 5U and 25 conformed with the Class C criterion for BOD in July and August respectively.
- In July and September, no sampling was carried out in Jala-jala River (Stn.20) because the river was dry.
- For the 3rd quarter, the BOD concentrations ranged from 1.0 to 281 mg/L with the highest concentration measured in Tunasan River -Downstream (Stn. 5) in July.

B. On Dissolved Oxygen (D.O.):

Laguna Lake:

- From July to September, all of the nine (9) lake stations conformed with the DENR Class C criterion for D.O. set at a minimum 5 mg/L except Stn. XV (4.0mg/L) in July. In Stn. XV, the diffusion of oxygen into the water could have been affected by the high temperature of the water since oxygen tends to be less soluble as temperature increases.
- The highest D.O. concentration at 15.9 mg/L was noted in Stn.IV in July.

Tributary Rivers:

- For the 3rd quarter of 2016, out of the 36 tributary stations monitored, only 7 tributary stations consistently conformed with the Class C criterion for DO and these were Stns. 8U, 13, 16, 17, 17U, 19 and 19U.
- The monthly DO concentrations in the fifteen (15) tributary stations consistently failed the 5.0 mg/L Class C criterion and these were Stns. 2, 3, 4, 5, 6, 7, 8, 8M, 9, 10, 14, 18, 24U, 25, and 26.
- The DO concentrations in Stns. 4U and 11 passed the criterion in July and September; Stns. 5U, 22B and 22C in July and August; Stns. 12 and 24 in September; and Stns. 1, 15, 21, 22A, 23 and 27 in August.
- The monthly DO concentrations in tributaries ranged from 0.05 to 8.9 mg/L.

C. **On Total Coliform (T. Coli) :**

Laguna Lake:

- For the 3rd quarter of 2016, the monthly geomeans for total coliform in all of the nine (9) lake stations conformed with the DENR Class C criterion of 5000 MPN/100ml.
- The highest monthly geomean for T.coli was noted in Stn. V with a value of 945 MPN/100mL in August while the lowest monthly geomean was noted in XVIII with geomean value of 38 MPN/100mL in July.

Tributary Rivers:

- From July to September, all of the the monthly geomean T. coli concentrations in all the tributary stations failed the Class C criterion.
- The highest monthly geomean of 80,166,321 MPN/100ml was recorded in San Cristobal River (Stn.10) in September while the lowest monthly geomean of 9,327 MPN/100mL was recorded in Tanay River-Daranak Falls (Stn.22B).

D. **On pH :**

Laguna Lake:

- The DENR Class C criteria for pH is 6.5 to 8.5.
- For the 3rd quarter of 2016, the recorded pH values in the lake ranged from 7.8 to 9.8. Six (6) stations consistently exceeded the maximum criterion of 8.5 namely; Stns. II, IV, VIII, XVI, XVII and XVIII.
- High pH values could be due to high photosynthetic activity of algae at the time of sampling. Deeper light penetration brought about by the clearing of the lake due to Pasig River Backflow could have also triggered increase in algal production in the lake.

Tributary Rivers:

- From July to September, all tributary stations passed the 6.5 to 8.5 Class C criteria for pH except in Stn. 10 (San Cristobal River) in August and Stn. 19U in July with recorded pH at 6.1 and 8.6, respectively..

E. **On Ammonia:**

Laguna Lake:

- The ammonia concentrations in the entire lake ranged from 0.001 to 0.313 mg/L.
- DAO 34 has no Class C criterion for ammonia.

F. **On Nitrate:**

Laguna Lake:

- For the 3rd quarter of 2016, all the nine (9) lake stations conformed with the DENR Class C criterion set at 10 mg/L.
- The nitrate concentrations in all of the nine (9) lake stations ranged from 0.001 to 5.052 mg/L.

Tributary Rivers:

- For the third quarter, the nitrate concentrations in all 35 tributary stations conformed with the 10 mg/L Class C criterion for nitrate.
- Measured nitrate concentrations in the tributary stations ranged from 0.001mg/L to 5.052mg/L. The highest nitrate concentration was noted in San Juan River (Stn. 11) in August.

G. On Inorganic Phosphate:

Laguna Lake:

- From July to September, out of the nine (9) lake stations, only two (2) lake stations consistently conformed with the 0.1 mg/L DENR Class C criterion and these were Stns. XVI and XVIII.
- The phosphate concentrations in Stn.I always failed to meet the Class C criterion.
- The highest inorganic phosphate was at 0.306 mg/L in Stn. V in August while the lowest inorganic concentration was at 0.001 mg/L in Stns. II , VIII and XVIII in August.

Tributary Rivers:

- For the tributary rivers the DENR Class C criterion for Inorganic Phosphate is set at 0.4 mg/L.Out of 36 monitored stations, only eleven (11) tributary river stations consistently conformed with the criterion and these were Stns. 1, 14 to 18, 19U, 22A, 22B,and 22C.
- Ten (10) tributary stations always failed the criterion, namely; Stns. 2, 3, 5, 5U, 6, 11, 23, 24, 24U and 26.
- The levels of inorganic phosphate in tributaries ranged from 0.002 mg/L to 4.016mg/L.

Comparatively, the lake water quality in the 2nd quarter was slightly better than in the 3rd quarter based on measured pH and Inorganic Phosphate concentrations.

For pH, only 1 station (Stn. XVIII) consistently failed the criterion in the 2nd quarter while six (6) stations consistently failed the criterion in 3rd quarter. The inorganic phosphate concentration in the 2nd quarter is better than in 3rd quarter because three (3) stations consistently conformed with 0.1 mg/L Class C criterion for inorganic phosphate while in 3rd quarter, only two (3) stations consistently conformed with criterion.

In terms of BOD, the water quality in the tributaries was better in the 3rd quarter than in the 2nd quarter because more stations conformed with Class C criterion. Seventeen (17) tributary stations consistently conformed with the BOD criterion as compared to only fifteen (15) tributaries in 2nd quarter.